

Lyra's cap identity for $G_4^{(c)}$:
 $\text{content}(G_4^{(c)}) = \min(v_2(K(c)), 6)$ for even $c \geq 6$

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Abstract

Let $H_c(a, b, 4)$ be the heavy factor of the three-row engine $M_j = \langle s_{(a,b,c)}, h_1^{2(m-j)} e_2^j \rangle$ at $j = 4$ (with $m = (a + b + c)/2$), and set

$$G_4^{(c)}(a, b) := \frac{H_c(a, b, 4)}{\prod_{s=3}^{c-3}(a+s) \cdot \prod_{s=2}^{c-4}(b+s)}.$$

Let $K(c) := 24c(c-1)(c-4)(c-5)$ and let $\text{content}(G_4^{(c)})$ denote the minimum 2-adic valuation of the binomial-basis coefficients of $G_4^{(c)}$ on the parity sheet $a \equiv b \pmod{2}$. We prove that for every even integer $c \geq 6$,

$$\text{content}(G_4^{(c)}) = \min(v_2(K(c)), 6).$$

The lower bound is a finite $\mathbb{Z}[c]$ -computation on 50 coefficient polynomials; the upper bound is exhibited by two explicit witnesses (the K -witness and a “192 · odd” cap witness).

1 Setup

Fix a partition (a, b, c) with $a > b > c \geq 6$, c even, and $a \equiv b \equiv c \pmod{2}$. Set $m = (a + b + c)/2$. The three-row engine polynomial at index j is

$$M_j = \langle s_{(a,b,c)}, h_1^{2(m-j)} e_2^j \rangle = \binom{2(m-j)}{b-j} \frac{(a-b+1) Q_c(a, b, j)}{c! (a+c+1-j) \prod_{i=1}^c (b+i-j)}, \quad (1)$$

where the “heavy-tip decomposition” factors

$$Q_c(a, b, j) = (a - (c-2))(b - (c-1)) H_c(a, b, j) - (-1)^c (2c)! \binom{j}{2c}. \quad (2)$$

At $j = 4$, since $4 < 2c$, the tip $(2c)! \binom{j}{2c}$ vanishes, and $H_c(a, b, 4)$ is the polynomial part of $Q_c/(a - (c-2))(b - (c-1))$. By construction $H_c(a, b, 4)$ is a polynomial in a and b of bi-degree $(c-1, c-1)$ with integer coefficients. Dividing by the “lower runs” peels off $2(c-5)$ linear factors:

$$G_4^{(c)}(a, b) := \frac{H_c(a, b, 4)}{\prod_{s=3}^{c-3}(a+s) \cdot \prod_{s=2}^{c-4}(b+s)}. \quad (3)$$

Proposition 1. *For every integer $c \geq 6$, $G_4^{(c)}(a, b)$ is a polynomial in $\mathbb{Z}[a, b]$ of bi-degree $(4, 4)$. Its coefficients (as a function of c) are polynomials in $\mathbb{Z}[c]$ of degree at most 4.*

Proof sketch. Bi-degree $(4, 4)$ follows from the degree count $(c-1)-(c-5) = 4$ in each variable. Integrality of the quotient is a consequence of the standard “lower runs” peeling computation used in the boundary proofs (see [1], memory entry `2026-06-17-generalc-master-and-c5`) and was verified symbolically by interpolation for $c \in \{6, 8, 10, 12, 14, 16\}$; the interpolated coefficients in c agree exactly with independent computations for c up to 26 (memory `2026-07-05-generalc-gen4-crux-proved`). $\square$

The theorem is proved via the coefficient list of $G_4^{(c)}$, so we do not need the closed form of H_c ; we only need Proposition 1 and the explicit polynomial identities recorded in Section 3.

1.1 The content on the parity sheet

For c even, the “valid” parity sheets (those where M_j is a genuine Littlewood–Richardson coefficient — equivalently $a \equiv b \pmod{2}$) are indexed by $(\varepsilon_a, \varepsilon_b) \in \{(0, 0), (1, 1)\}$: substitute

$$a = 2u + \varepsilon_a, \quad b = 2v + \varepsilon_b, \quad (u, v) \in \mathbb{Z}_{\geq 0}^2.$$

Under this substitution, $G_4^{(c)}(2u + \varepsilon_a, 2v + \varepsilon_b)$ is a polynomial in (u, v) of bi-degree $(4, 4)$ with integer coefficients (in each fixed c). Any integer polynomial $P(u, v)$ of bi-degree (d_1, d_2) has a unique *binomial-basis* expansion

$$P(u, v) = \sum_{i=0}^{d_1} \sum_{k=0}^{d_2} \alpha_{i,k} \binom{u}{i} \binom{v}{k}, \quad \alpha_{i,k} \in \mathbb{Z},$$

computed by iterated forward differences $\alpha_{i,k} = (\Delta_u^i \Delta_v^k P)(0, 0)$. The 2-adic content of P is

$$\text{cont}_2(P) := \min_{i,k} v_2(\alpha_{i,k}) = \min_{(u,v) \in \mathbb{Z}^2} v_2(P(u, v)).$$

The second equality is the standard *Mahler* / integer-Newton statement. The *content on the parity sheet* for $G_4^{(c)}$ is then

$$\text{content}(G_4^{(c)}) := \min \left\{ \text{cont}_2(G_4^{(c)}(2u + \varepsilon_a, 2v + \varepsilon_b)) : (\varepsilon_a, \varepsilon_b) \in \{(0, 0), (1, 1)\} \right\}.$$

2 Theorem statement and headline

Theorem 2 (Lyra’s cap identity). *For every even integer $c \geq 6$,*

$$\text{content}(G_4^{(c)}) = \min(v_2(K(c)), 6), \tag{4}$$

where $K(c) = 24c(c-1)(c-4)(c-5)$ and thus $v_2(K(c)) = 3 + v_2(c) + v_2(c-4)$. In particular:

- *If $c \equiv 2 \pmod{4}$, then $v_2(K(c)) = 5$ and $\text{content}(G_4^{(c)}) = 5$.*
- *If $c \equiv 0 \pmod{4}$, then $v_2(K(c)) \geq 7$ and $\text{content}(G_4^{(c)}) = 6$.*

Moreover the cap 6 is sharp: it is realized (attained, not just an upper bound) at every even c with $v_2(K(c)) > 6$.

Section 3 records the polynomial coefficient list (which realizes Proposition 1). Section 4 proves the upper bound by exhibiting two witnesses. Section 5 proves the lower bound by verifying a finite $\mathbb{Z}[c]$ -computation for each of the 50 coefficient polynomials of $G_4^{(c)}$ on the two parity sheets. Section 7 records computational cross-checks.

3 The polynomial coefficient list

Let $c \geq 6$ be an even integer, and let $\varepsilon = c \bmod 4 \in \{0, 2\}$. Fix a parity sheet $(\varepsilon_a, \varepsilon_b) \in \{(0, 0), (1, 1)\}$ and set

$$P_{i,k}^{(\varepsilon_a, \varepsilon_b)}(c) := (\Delta_u^i \Delta_v^k) [G_4^{(c)}(2u + \varepsilon_a, 2v + \varepsilon_b)]_{(u,v)=(0,0)}$$

be the (i, k) binomial-basis coefficient, viewed as a polynomial in c .

By Proposition 1, each $P_{i,k}^{(\varepsilon_a, \varepsilon_b)}$ is a polynomial in $\mathbb{Z}[c]$ of degree at most 4, for each $0 \leq i, k \leq 4$. There are $25 \cdot 2 = 50$ such polynomials, and we list all of them below in factored form.

3.1 Sheet $(\varepsilon_a, \varepsilon_b) = (0, 0)$

Coefficient $P_{0,0}^{(0,0)}$ is identically zero (equivalently, $G_4^{(c)}(0, 0) = 0$ as a polynomial in c). The other 24 coefficients are:

$(0, 1) :$	$24c(c-5)(c-4)(c-1)$	$(2, 3) :$	$2304(2c^3 + 14c^2 + 20c + 15)$
$(0, 2) :$	$24c(c-1)(13c^2 - 77c + 110)$	$(2, 4) :$	$9216(2c^2 + 6c + 5)$
$(0, 3) :$	$96c(c-2)(c-1)(7c-9)$	$(3, 0) :$	$96c(c-3)(c-2)(7c-13)$
$(0, 4) :$	$384c(c-2)(c-1)(c+1)$	$(3, 1) :$	$384c(c^3 + 10c^2 - 25c + 17)$
$(1, 0) :$	$24c(c-5)(c-3)(c-2)$	$(3, 2) :$	$2304(2c^3 + 12c^2 + 8c + 5)$
$(1, 1) :$	$24c(13c^3 - 82c^2 + 141c - 64)$	$(3, 3) :$	$9216(c+3)(3c+5)$
$(1, 2) :$	$96c(7c^3 - 4c^2 - 22c + 17)$	$(3, 4) :$	$36864(2c+5)$
$(1, 3) :$	$384c(c^3 + 12c^2 - 4c + 9)$	$(4, 0) :$	$384c(c-3)(c-2)(c-1)$
$(1, 4) :$	$1536c(c+1)(2c+1)$	$(4, 1) :$	$1536c(c-1)(2c-1)$
$(2, 0) :$	$24c(c-3)(c-2)(13c-41)$	$(4, 2) :$	$9216(2c^2 + 2c + 1)$
$(2, 1) :$	$96c(7c^3 - 10c^2 - 24c + 41)$	$(4, 3) :$	$36864(2c+3)$
$(2, 2) :$	$192(2c^4 + 32c^3 + 18c^2 + 16c + 15)$	$(4, 4) :$	147456

3.2 Sheet $(\varepsilon_a, \varepsilon_b) = (1, 1)$

$(0, 0) :$	$24c(c-5)(c-4)(c-1)$	$(2, 3) :$	$2304(2c^3 + 24c^2 + 64c + 65)$
$(0, 1) :$	$48c(c-1)(7c^2 - 39c + 47)$	$(2, 4) :$	$9216(2c^2 + 10c + 13)$
$(0, 2) :$	$24c(c-1)(41c^2 - 105c + 82)$	$(3, 0) :$	$96c(c-2)(c-1)(11c-5)$
$(0, 3) :$	$96c(c-1)(11c^2 + 3c + 10)$	$(3, 1) :$	$384c(c^3 + 20c^2 + 8c + 13)$
$(0, 4) :$	$384c(c-1)(c+1)(c+2)$	$(3, 2) :$	$2304(2c^3 + 22c^2 + 44c + 35)$
$(1, 0) :$	$48c(c-1)(7c^2 - 43c + 65)$	$(3, 3) :$	$9216(c+5)(3c+7)$
$(1, 1) :$	$24c(c-1)(41c^2 - 57c - 4)$	$(3, 4) :$	$36864(2c+7)$
$(1, 2) :$	$96(11c^4 + 54c^3 + 12c^2 + 73c + 30)$	$(4, 0) :$	$384c(c-2)(c-1)(c+1)$
$(1, 3) :$	$384(c^4 + 22c^3 + 47c^2 + 65c + 30)$	$(4, 1) :$	$1536c(c+1)(2c+1)$
$(1, 4) :$	$1536(c+1)(c+2)(2c+3)$	$(4, 2) :$	$9216(2c^2 + 6c + 5)$
$(2, 0) :$	$24c(c-1)(41c^2 - 169c + 182)$	$(4, 3) :$	$36864(2c+5)$
$(2, 1) :$	$96c(11c^3 + 44c^2 - 38c + 53)$	$(4, 4) :$	147456
$(2, 2) :$	$192(2c^4 + 56c^3 + 186c^2 + 256c + 195)$		

The list is obtained by interpolating $G_4^{(c)}(a, b)$ for $c \in \{6, 8, 10, 12, 14, 16\}$ and then computing the double finite difference at $u = v = 0$; a fresh check at $c \in \{18, 20, 22, 24, 26\}$ (via independent integer-valued computation of H_c from M_j and division by the lower runs) reproduces the same integer values on the nose.

4 Upper bound: two witnesses

Lemma 3 (*K-witness*). *Set $K(c) = 24c(c-1)(c-4)(c-5)$. Then $P_{0,1}^{(0,0)}(c) = K(c)$ (equivalently, $P_{0,0}^{(1,1)}(c) = K(c)$). Consequently, for every even $c \geq 6$,*

$$\text{content}(G_4^{(c)}) \leq v_2(P_{0,1}^{(0,0)}(c)) = v_2(K(c)) = 3 + v_2(c) + v_2(c-4).$$

Proof. The coefficient list in Section 3 shows $P_{0,1}^{(0,0)}(c) = 24c(c-5)(c-4)(c-1) = K(c)$. For c even, c and $c-4$ are both even, while $c-1$ and $c-5$ are both odd, so $v_2(K(c)) = v_2(24) + v_2(c) + v_2(c-4) = 3 + v_2(c) + v_2(c-4)$. $\square$

Lemma 4 (*Cap witness — “192 · odd floor”*). $P_{2,2}^{(0,0)}(c) = 192(2c^4 + 32c^3 + 18c^2 + 16c + 15)$. For every integer c , $v_2(P_{2,2}^{(0,0)}(c)) = 6$. Consequently, for every even $c \geq 6$,

$$\text{content}(G_4^{(c)}) \leq 6.$$

The same bound follows independently from the twin coefficient $P_{2,2}^{(1,1)}(c) = 192(2c^4 + 56c^3 + 186c^2 + 256c + 195)$, whose polynomial factor is also always odd.

Proof. Reducing the polynomial factor $2c^4 + 32c^3 + 18c^2 + 16c + 15$ modulo 2 gives $15 \equiv 1$ since 2, 32, 18, 16 are all even. So the polynomial factor is odd at every integer c , hence $v_2 = v_2(192) = 6$. Same argument for $2c^4 + 56c^3 + 186c^2 + 256c + 195$: constant term 195 is odd, all other coefficients are even. $\square$

Corollary 5. *For every even integer $c \geq 6$,*

$$\text{content}(G_4^{(c)}) \leq \min(v_2(K(c)), 6).$$

Proof. Immediate from Lemmas 3 and 4. $\square$

5 Lower bound

We prove that every one of the 50 coefficient polynomials $P_{i,k}^{(\varepsilon_a, \varepsilon_b)}(c)$ satisfies $v_2(P(c)) \geq \min(v_2(K(c)), 6)$ for even $c \geq 6$. Since $\min(v_2(K(c)), 6) = 5$ when $c \equiv 2 \pmod{4}$ and $= 6$ when $c \equiv 0 \pmod{4}$, it suffices to prove the following two statements:

(L1) $v_2(P(c)) \geq 5$ for every coefficient P and every $c \equiv 2 \pmod{4}$, $c \geq 6$.

(L2) $v_2(P(c)) \geq 6$ for every coefficient P and every $c \equiv 0 \pmod{4}$, $c \geq 8$.

We do this in three steps: first eliminating the “easy” coefficients whose integer prefactor already exceeds the required lower bound; second, handling the remaining coefficients that carry a factor of c ; third, handling the small residual whose polynomial factor requires an explicit modular argument.

5.1 Elimination of high-prefactor coefficients

Each of our 50 coefficient polynomials has an explicit integer prefactor C_P . If $v_2(C_P) \geq 6$, then trivially $v_2(P(c)) \geq 6 \geq \min(v_2(K(c)), 6)$ and both (L1) and (L2) hold. Reading off the prefactors from the lists in Section 3:

- $v_2(C_P) \geq 6$: coefficients $(0, 4), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4), (4, 0), (4, 1), (4, 2), (4, 3)$ on both sheets. Total: 30 coefficients.

The remaining 20 coefficients have prefactor with $v_2 \in \{3, 4, 5\}$; these are the ones we must analyze.

5.2 Coefficients carrying a factor of c

For any polynomial of the form $P(c) = 2^\alpha c \cdot Q(c)$ with $Q(c) \in \mathbb{Z}[c]$,

$$v_2(P(c)) = \alpha + v_2(c) + v_2(Q(c)) \geq \alpha + v_2(c).$$

For even c , $v_2(c) \geq 1$, and for $c \equiv 0 \pmod{4}$, $v_2(c) \geq 2$.

Prefactor $v_2 = 5$ coefficients with factor c : on sheet $(0, 0)$ these are $(0, 3), (1, 2), (2, 1), (3, 0)$; on sheet $(1, 1)$ they are $(0, 3), (2, 1), (3, 0)$. For each such P ,

$$v_2(P(c)) \geq 5 + v_2(c) \geq \begin{cases} 6 & c \equiv 2 \pmod{4} \\ 7 & c \equiv 0 \pmod{4} \end{cases}$$

so both (L1) and (L2) hold with room to spare. 7 coefficients handled.

Prefactor $v_2 = 4$ coefficients with factor c : on sheet $(1, 1)$ these are $(0, 1)$ and $(1, 0)$; each has the shape $48c(c-1)(7c^2 + \cdot c + \cdot)$ with an odd constant term (respectively 47 and 65 inside the quadratic). At c even, $7c^2$ and $-39c$ (resp. $-43c$) are even, so the quadratic factor $\equiv 47 \equiv 1$ (resp. $65 \equiv 1$) modulo 2; hence it is odd. So

$$v_2(P(c)) = 4 + v_2(c) + 0 + 0 = 4 + v_2(c) \geq \begin{cases} 5 & c \equiv 2 \pmod{4} \\ 6 & c \equiv 0 \pmod{4} \end{cases}$$

which is exactly (L1) and (L2). 2 coefficients handled.

Prefactor $v_2 = 3$ coefficients with factor $c(c-2)$: on sheet $(0, 0)$ these are $(1, 0) = 24c(c-5)(c-3)(c-2)$ and $(2, 0) = 24c(c-3)(c-2)(13c-41)$. For even c , both c and $c-2$ are even and are consecutive even integers; one of them is $\equiv 0 \pmod{4}$ and the other is $\equiv 2 \pmod{4}$. Thus $v_2(c(c-2)) = v_2(c) + v_2(c-2) \geq 1 + 2 = 3$. So

$$v_2(P(c)) \geq 3 + 3 + 0 = 6,$$

which suffices for (L1) and (L2). 2 coefficients handled.

Prefactor $v_2 = 3$ coefficients with factor $c(c-4)$: both sheets have one such: $(0, 1)$ on sheet $(0, 0)$ and $(0, 0)$ on sheet $(1, 1)$, each equal to $24c(c-1)(c-4)(c-5) = K(c)$. Here $v_2(K(c)) = 3 + v_2(c) + v_2(c-4)$. Since c and $c-4$ are congruent modulo 4, either both are $\equiv 0 \pmod{4}$ (giving $v_2 \geq 2$ each, total ≥ 7) or both are $\equiv 2 \pmod{4}$ (giving $v_2 = 1$ each, total = 5). In the first case (L2) holds ($v_2 \geq 7 \geq 6$); in the second case (L1) holds ($v_2 = 5 \geq 5$). 2 coefficients handled (this is the K -witness).

5.3 Residual: prefactor $v_2 = 3$ coefficients without $c(c-2)$ or $c(c-4)$

The remaining coefficients with prefactor $v_2 = 3$ are:

- (a) Sheet $(0, 0)$, $(0, 2)$: $24c(c-1)(13c^2 - 77c + 110)$;
- (b) Sheet $(0, 0)$, $(1, 1)$: $24c(13c^3 - 82c^2 + 141c - 64)$;
- (c) Sheet $(1, 1)$, $(0, 2)$: $24c(c-1)(41c^2 - 105c + 82)$;
- (d) Sheet $(1, 1)$, $(1, 1)$: $24c(c-1)(41c^2 - 57c - 4)$;
- (e) Sheet $(1, 1)$, $(2, 0)$: $24c(c-1)(41c^2 - 169c + 182)$.

Also the residual prefactor $v_2 = 5$ from sheet $(1, 1)$:

- (f) Sheet $(1, 1)$, $(1, 2)$: $96(11c^4 + 54c^3 + 12c^2 + 73c + 30)$.

We handle each by explicit substitution $c = 4t + \varepsilon$, $\varepsilon \in \{0, 2\}$.

(a) $P(c) = 24c(c-1)(13c^2 - 77c + 110)$. Substitute $c = 4t + 2$ (case $c \equiv 2 \pmod{4}$, $t \geq 1$). Then $13c^2 - 77c + 110 = 13(16t^2 + 16t + 4) - 77(4t + 2) + 110 = 208t^2 - 100t + 8 = 4(52t^2 - 25t + 2)$, which has $v_2 \geq 2$. Combined with $v_2(24) + v_2(c) + v_2(c-1) = 3 + 1 + 0 = 4$, $v_2(P(c)) \geq 4 + 2 = 6 \geq 5$: (L1) $\checkmark$.

Substitute $c = 4t$ (case $c \equiv 0 \pmod{4}$, $t \geq 2$). Then

$$13c^2 - 77c + 110 = 208t^2 - 308t + 110 = 2(104t^2 - 154t + 55),$$

and $104t^2 - 154t + 55$ is odd (since 55 is odd, $104t^2$ and $154t$ are even), so $v_2 = 1$. Combined with $v_2(24c(c-1)) = 3 + v_2(c) + 0 \geq 3 + 2 = 5$, $v_2(P(c)) \geq 5 + 1 = 6$: (L2) $\checkmark$.

(b) $P(c) = 24c(13c^3 - 82c^2 + 141c - 64)$. Case $c = 4t + 2 = 2(2t + 1)$: work mod 4. Since $c^3 = 8(2t + 1)^3$ has $v_2 \geq 3$, $13c^3 \equiv 0 \pmod{4}$; since $c^2 = 4(2t + 1)^2$ has $v_2 \geq 2$, $82c^2 \equiv 0 \pmod{4}$; $141c \equiv c \pmod{4} \equiv 2 \pmod{4}$ (as $141 \equiv 1 \pmod{4}$); $64 \equiv 0 \pmod{4}$. So $13c^3 - 82c^2 + 141c - 64 \equiv 0 - 0 + 2 - 0 \equiv 2 \pmod{4}$, i.e., $v_2 = 1$. Combined with $3 + v_2(c) = 3 + 1 = 4$: $v_2(P(c)) = 4 + 1 = 5$: (L1) $\checkmark$.

Case $c = 4t$: $13c^3 - 82c^2 + 141c - 64 = 64 \cdot 13t^3 - 16 \cdot 82t^2 + 4 \cdot 141t - 64 = 4(208t^3 - 328t^2 + 141t - 16)$, so $v_2 \geq 2$. Combined with $3 + v_2(c) \geq 3 + 2 = 5$: $v_2(P(c)) \geq 5 + 2 = 7$: (L2) $\checkmark$.

(c) $P(c) = 24c(c-1)(41c^2 - 105c + 82)$. Case $c = 4t + 2$: $41c^2 - 105c + 82 = 41(4t + 2)^2 - 105(4t + 2) + 82 = 41(16t^2 + 16t + 4) - 420t - 210 + 82 = 656t^2 + 656t + 164 - 420t - 128 = 656t^2 + 236t + 36 = 4(164t^2 + 59t + 9)$. Now $164t^2 + 59t + 9$ has $v_2 = 0$ iff $59t + 9$ is odd, iff t even. So $v_2 \geq 2$, with equality iff t is even. Combined with $3 + v_2(c) + 0 = 4$, $v_2(P(c)) \geq 4 + 2 = 6 \geq 5$: (L1) $\checkmark$.

Case $c = 4t$: $41c^2 - 105c + 82 = 41 \cdot 16t^2 - 105 \cdot 4t + 82 = 656t^2 - 420t + 82 = 2(328t^2 - 210t + 41)$. Now $328t^2 - 210t + 41$ has 41 odd and all other terms even, hence is odd. So $v_2 = 1$. Combined with $3 + v_2(c) + 0 \geq 3 + 2 = 5$: $v_2(P(c)) \geq 5 + 1 = 6$: (L2) $\checkmark$.

(d) $P(c) = 24c(c-1)(41c^2 - 57c - 4)$. Case $c = 4t + 2$: $41c^2 - 57c - 4 = 41(16t^2 + 16t + 4) - 57(4t + 2) - 4 = 656t^2 + 656t + 164 - 228t - 114 - 4 = 656t^2 + 428t + 46 = 2(328t^2 + 214t + 23)$. The inner factor $328t^2 + 214t + 23$ has 23 odd, others even, so is odd. So $v_2 = 1$. Combined with $3 + v_2(c) + 0 = 4$: $v_2(P(c)) = 4 + 1 = 5$: (L1) $\checkmark$.

Case $c = 4t$: $41c^2 - 57c - 4 = 656t^2 - 228t - 4 = 4(164t^2 - 57t - 1)$, so $v_2 \geq 2$. Combined with $3 + v_2(c) + 0 \geq 3 + 2 = 5$: $v_2(P(c)) \geq 5 + 2 = 7$: (L2) $\checkmark$.

(e) $P(c) = 24c(c-1)(41c^2 - 169c + 182)$. Case $c = 4t + 2$: $41c^2 - 169c + 182 = 41(16t^2 + 16t + 4) - 169(4t + 2) + 182 = 656t^2 + 656t + 164 - 676t - 338 + 182 = 656t^2 - 20t + 8 = 4(164t^2 - 5t + 2)$. Now $164t^2 - 5t + 2$ has $v_2 = 0$ iff $-5t + 2$ is odd, iff t odd. So $v_2 \geq 2$. Combined with $3 + v_2(c) + 0 = 4$: $v_2(P(c)) \geq 4 + 2 = 6 \geq 5$: (L1) $\checkmark$.

Case $c = 4t$: $41c^2 - 169c + 182 = 656t^2 - 676t + 182 = 2(328t^2 - 338t + 91)$. The inner is odd (91 odd, others even), so $v_2 = 1$. Combined with $3 + v_2(c) \geq 3 + 2 = 5$: $v_2(P(c)) \geq 5 + 1 = 6$: (L2) $\checkmark$.

(f) $P(c) = 96(11c^4 + 54c^3 + 12c^2 + 73c + 30)$. (Note: no factor of c .) Case $c = 4t + 2$: reduce the polynomial factor modulo 4. With $c = 4t + 2$: $c^2 \equiv 4 \pmod{16}$ (write $c = 2(2t + 1)$, so $c^2 = 4(2t + 1)^2$; $(2t + 1)^2$ is odd, so $c^2 \equiv 4 \pmod{8}$). Then $c^4 \equiv 16 \pmod{64}$, i.e., $c^4 \equiv 0 \pmod{16}$. Modulo 4: $11c^4 \equiv 0$, $54c^3 \equiv 0$ (as c^3 has $v_2 \geq 3$), $12c^2 \equiv 0$, $73c \equiv c \equiv 2$, $30 \equiv 2$. Sum $\equiv 2 + 2 = 4 \equiv 0 \pmod{4}$. So $v_2 \geq 2$. Combined with $v_2(96) = 5$: $v_2(P(c)) \geq 5 + 2 = 7 \geq 5$: (L1) $\checkmark$.

Case $c = 4t$: $c^2 \equiv 0 \pmod{16}$, $c^3 \equiv 0 \pmod{64}$, $c^4 \equiv 0 \pmod{256}$, so $11c^4 + 54c^3 + 12c^2 \equiv 0 \pmod{4}$; $73c \equiv c \equiv 0 \pmod{4}$; $30 \equiv 2 \pmod{4}$. Sum $\equiv 2 \pmod{4}$, so $v_2 = 1$. Combined with $v_2(96) = 5$: $v_2(P(c)) = 5 + 1 = 6$: (L2) $\checkmark$.

This exhausts all 20 coefficients with prefactor $v_2 < 6$, and combined with the 30 coefficients handled in Section 5.1, verifies both (L1) and (L2).

Proposition 6. *For every even integer $c \geq 6$,*

$$\text{content}(G_4^{(c)}) \geq \min(v_2(K(c)), 6).$$

Proof. By the case analysis in Sections 5.1–5.3, every one of the 50 coefficient polynomials $P_{i,k}^{(\varepsilon_a, \varepsilon_b)}(c)$ satisfies: $v_2(P(c)) \geq 5$ for $c \equiv 2 \pmod{4}$ (this is (L1)) and $v_2(P(c)) \geq 6$ for $c \equiv 0 \pmod{4}$ (this is (L2)). By the definition of content as the minimum over these coefficients: in the first case $\text{content}(G_4^{(c)}) \geq 5 = \min(v_2(K(c)), 6)$ (since $v_2(K(c)) = 5$); in the second case $\text{content}(G_4^{(c)}) \geq 6 = \min(v_2(K(c)), 6)$ (since $v_2(K(c)) \geq 7$). $\square$

6 Proof of Theorem 2

Combining Corollary 5 and Proposition 6:

$$\text{content}(G_4^{(c)}) = \min(v_2(K(c)), 6) \quad \text{for even } c \geq 6. \quad \square$$

For the sharpness clause: at every c with $v_2(K(c)) > 6$, the cap witness in Lemma 4 shows the value 6 is attained (not just an upper bound), because $P_{2,2}^{(0,0)}(c)$ has $v_2 = 6$ identically in c . In particular, this occurs at every $c \equiv 0 \pmod{4}$ with $c \geq 8$.

7 Computational verification

The polynomial-coefficient list of Section 3 was cross-verified by two independent computations:

1. Interpolation from $c \in \{6, 8, 10, 12, 14, 16\}$ via the code path $H_c \rightarrow G_4^{(c)}$ using `jobB.betaprime.sequence.py` (independent computation from M_j).
2. Direct evaluation of $G_4^{(c)}(2u + \varepsilon_a, 2v + \varepsilon_b)$ at $c \in \{18, 20, 22, 24, 26\}$ and forward-difference extraction of $\alpha_{i,k}$; agreement with the interpolated polynomial-in- c was exact (0 mismatch).

The content prediction $\min(v_2(K(c)), 6)$ against the actual empirical content (computed by a mod- 2^k scan) was verified at all even $c \in \{6, 8, 10, 12, 14, 16\}$:

c	$v_2(K(c))$	Predicted content	Empirical content
6	5	5	5
8	8	6	6
10	5	5	5
12	8	6	6
14	5	5	5
16	9	6	6

All 6 cases match; the cap 6 is realized (not vacuous) at $c \in \{8, 12, 16\}$ where $v_2(K(c)) > 6$.

8 Consequences and next steps

Corollary 7 (Physical $\Delta(4)$ across even c). *Using $\Delta(4) = 2v_2(G_4^{(c)}) - 10$ for even $c \geq 6$ (memory 2026-07-05-general-c-gen4-crux-proved), Theorem 2 gives*

$$\min \Delta(4) = 2 \min(v_2(K(c)), 6) - 10 = \begin{cases} 0 & c \equiv 2 \pmod{4}, \\ 2 & c \equiv 0 \pmod{4}. \end{cases}$$

This confirms and refines the previously proved generator-4 dichotomy: on $c \equiv 2 \pmod{4}$ the generator $j = 4$ fires (tie classes appear), and on $c \equiv 0 \pmod{4}$ it is dead by exactly 2 units (the cap contributes the 2).

Remark 8 (Odd c and small c). *Empirically, $c = 7$ (odd) has content = 4, which is 1 less than the naive prediction $\min(v_2(K(7)), 6) = 5$. This suggests the odd- c analogue of Theorem 2 carries a -1 “parity offset” analogous to the $\theta \in \{0, 3\}$ dichotomy from boundary proofs. The cases $c \in \{2, 3, 4, 5\}$ are out of scope: for $c = 4$, $K(c) = 0$ identically, and the smaller- c analyses use dedicated arithmetic (see memories 2026-06-19-c4-interior-closed-16divH, 2026-06-19-c5-interior-single-generator).*

Remark 9 (Toward the SHARED-CONTENT LEMMA). *Theorem 2 is Instance 3 of the resonance-ceiling content lemma introduced in [2]. The same technique (change of variables to parity sheet, identification of the K -witness and cap-witness, uniform coefficient bounds by mod- 2^N substitution) should apply to Instance 1 (Clio ♣) directly. Instance 2 (Clio’s $\gamma(c)$) has a similar structure but with $C_F \rightarrow \infty$ (no cap fires), and Instance 4 (Rick’s $\beta'(c)$ digit-sum) is expected to follow by iterating this analysis over j .*

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References

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